

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

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| 1. Step 1, E1: (diagram) $\rightarrow -3$ | 11. Step 4, E2: $(+6) \rightarrow = \text{opp. of } 6; = -6$ | 16. Step 6, E1: The opposite of a number is -14 . What is the number? $\rightarrow$ opposite of $? = -14$; the number is 14 |
| 2. Step 1, E2: (diagram) $\rightarrow 5$ | 12. Step 4, E3: $n = 15 \rightarrow -n = -(15); = -15$ | 17. Step 6, E2: A diver is 13 m below sea level. A gull is the same distance above. Write both heights as integers. $\rightarrow$ diver: -13 m; gull: 13 m |
| 3. Step 1, E3: 38 m above sea level $\rightarrow 38$ | 13. Step 5, E1: Which is further from 0: -8 or 9 ? $\rightarrow -8$: 8 units, 9 : 9 units; further: 9 | 18. Step 6, E3: Two integers are opposites and are 20 units apart. What are they? $\rightarrow$ each is half of 20 from 0: 10; 10 and -10 |
| 4. Step 1, E4: 39 floors down $\rightarrow -39$ | 14. Step 5, E2: Which two integers are 8 units from 0? $\rightarrow 8$ right of 0: 8; 8 left of 0: -8 | |
| 5. Step 2, E1: Thermometer A $\rightarrow -6^\circ\text{C}$ | 15. Step 5, E3: List every integer less than 4 units from 0. $\rightarrow$ from -3 up to 3; $-3, -2, -1, 0, 1, 2, 3$ | |
| 6. Step 2, E2: the kite $\rightarrow 20$ m | | |
| 7. Step 3, E1: The opposite of 4 $\rightarrow 4$ is 4 units from 0; opposite: -4 | | |
| 8. Step 3, E2: $57 \rightarrow -57$ | | |
| 9. Step 3, E3: 128 $\rightarrow -128$ | | |
| 10. Step 4, E1: $(-33) \rightarrow = \text{opp. of } -33; = 33$ | | |

Questions

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|------------------------|-------------|-----------------------|
| 1. -9 | 14. 15 m | 27. -20 |
| 2. 7 | 15. 0 m | 28. 28 |
| 3. -7 | 16. -15 m | 29. 30 |
| 4. -6 | 17. -20 m | 30. -7 |
| 5. 8 | 18. -5 | 31. 7 and -7 |
| 6. -23 | 19. -7 | 32. $-2, -1, 0, 1, 2$ |
| 7. 72 | 20. 65 | 33. 3 |
| 8. -85 | 21. -111 | 34. 9 and -9 |
| 9. 37 | 22. 60 | 35. 17 |
| 10. 43 | 23. -23 | 36. -6 m, 6 m |
| 11. 4°C | 24. 24 | 37. 15 and -15 |
| 12. 3°C | 25. -27 | |
| 13. -4°C | 26. 22 | |

1.02 The number line (booklet p. 5)

WE DO examples (teacher)

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|---|---|--|
| 1. Step 1, E1: (diagram) $\rightarrow 4$ | 6. Step 3, E1: What number is at the ? $\rightarrow 3$ gaps = 15, so step = 5; ? = 10 | 10. Step 5, E2: Halfway between -9 and 7 $\rightarrow 16$ apart, half is 8; $-9 + 8 = -1$ |
| 2. Step 1, E2: (diagram) $\rightarrow -2$ | 7. Step 4, E1: -5 and 1 $\rightarrow -5$ to 0: 5, 0 to 1: 1; $5 + 1 = 6$ | 11. Step 6, E1: How many integers are between -9 and 11? $\rightarrow -8$ to -1 : 8, and 0; 1 to 10: 10; $8 + 1 + 10 = 19$ |
| 3. Step 1, E3: -9 or $4 \rightarrow 4$ | 8. Step 4, E2: -14 and $-2 \rightarrow -14$ to -2 ; 12 units | |
| 4. Step 1, E4: 10 or $-9 \rightarrow 10$ | 9. Step 5, E1: -6 and 4 $\rightarrow 10$ apart, half is 5; $-6 + 5 = -1$ | |
| 5. Step 2, E1: Start at -6 and move 4 left. $\rightarrow$ land on -10 | | |

12. Step 6, E2: An ant starts at 6. It walks 12 units left, then 7 units right. Where is it now? $\rightarrow 6 - 12 = -6$; $-6 + 7 = 1$; it is at 1

13. Step 6, E3: Point A is at -10 . Point B is 8 units to the right of A. Where is B, and how far is B from 0? $\rightarrow -10 + 8 = -2$; B is at -2 ; 2 units from 0

Questions

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|---------|-----------|-----------|
| 1. 2 | 10. 14 | 19. 14 |
| 2. 7 | 11. -7 | 20. 11 |
| 3. 5 | 12. -10 | 21. 8 |
| 4. 1 | 13. -10 | 22. 0 |
| 5. 0 | 14. 9 | 23. -14 |
| 6. 2 | 15. 20 | 24. -14 |
| 7. -5 | 16. 25 | 25. 25 |
| 8. -8 | 17. 10 | 26. -1 |
| 9. -5 | 18. 0 | |

1.03 Comparing and ordering integers (booklet p. 7)

WE DO examples (teacher)

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| 1. Step 1, E1: -4 or $-2 \rightarrow -2$ | 10. Step 3, E2: 22, $-24, -7, -19, -31 \rightarrow$ largest: 22, smallest: -31 ; 22, $-7, -19, -24, -31$ | 14. Step 6, E1: Write three integers greater than -20 but less than -13 . $\rightarrow$ integers from -19 to -14 ; e.g. $-19, -18, -17$ |
| 2. Step 1, E2: -1 or $-2 \rightarrow -1$ | 11. Step 4, E1: Which city was the coldest? $\rightarrow$ lowest: -6°C ; Oslo | 15. Step 6, E2: How many integers are greater than -8 and less than 8? $\rightarrow -7$ up to 7; $7 + 1 + 7 = 15$ |
| 3. Step 1, E3: 3 or 6 $\rightarrow 3$ | 12. Step 5, E1: Which is the smallest: $-(-2)$, -9 or $-(-12)$? $\rightarrow -(-2) = 2, -(-12) = 12$; smallest: -9 | 16. Step 6, E3: I am an odd integer. I am less than 0 but greater than -8 . What could I be? $\rightarrow$ between -8 and 0; odd: $-7, -5, -3, -1$ |
| 4. Step 1, E4: -19 or $-30 \rightarrow -30$ | 13. Step 5, E2: Write in ascending order: $-(-4)$, $-8, -9, 0 \rightarrow -(-4) = 4; -9, -8, 0, 4$ | |
| 5. Step 2, E1: -12 $\square 0 \rightarrow <$ | | |
| 6. Step 2, E2: -19 $\square 10 \rightarrow <$ | | |
| 7. Step 2, E3: $-7 > -4 \rightarrow$ false | | |
| 8. Step 2, E4: $3 > 6 \rightarrow$ false | | |
| 9. Step 3, E1: 9, 2, $-1, -6 \rightarrow -6, -1, 2, 9$ | | |

Questions

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|----------|---------------------|----------------------------|
| 1. 1 | 13. $<$ | 25. 25, 3, $-15, -25, -28$ |
| 2. -2 | 14. $>$ | 26. 6, $-3, -8, -31, -39$ |
| 3. 1 | 15. $<$ | 27. Beijing |
| 4. -4 | 16. $<$ | 28. 2 |
| 5. -9 | 17. $>$ | 29. $-6, -4, 0, 5, 7$ |
| 6. -17 | 18. true | 30. Oslo, Denver, Toronto |
| 7. -13 | 19. true | 31. $>$ |
| 8. -17 | 20. true | 32. -3 |
| 9. 4 | 21. true | 33. $-3, -2, 0, 11$ |
| 10. $<$ | 22. false | |
| 11. $>$ | 23. $-4, -2, 1, 2$ | |
| 12. $>$ | 24. $-5, -4, -2, 3$ | |

34. -6 37. -12, -4, 0, 10 40. -7, -5, -3, -1
 35. -10, -2, 0, 9 38. e.g. -13, -12, 41. e.g. -18, -17,
 36. -6 -11 -16
 39. 15

1.04 Adding integers (booklet p. 9)

Step 1: Adding a positive

E1 (WE DO) $-2 + 4$
 $= 2$

1 $-4 + 3$
 $= -1$

2 $-5 + 2$
 $= -3$

3 $-6 + 4$
 $= -2$

4 $-3 + 7$
 $= 4$

5 $-7 + 7$
 $= 0$

Step 2: Adding a negative

E2 (WE DO) $6 + (-2)$
 $= 6 - 2 \cdot = 4$

E3 (WE DO) $-2 + (-3)$
 $= -2 - 3 \cdot = -5$

6 $7 + (-3)$
 $= 7 - 3 \cdot = 4$

7 $5 + (-5)$
 $= 5 - 5 \cdot = 0$

8 $2 + (-6)$
 $= 2 - 6 \cdot = -4$

9 $4 + (-10)$
 $= 4 - 10 \cdot = -6$

10 $-1 + (-4)$
 $= -1 - 4 \cdot = -5$

11 $-3 + (-6)$
 $= -3 - 6 \cdot = -9$

Step 3: Zero pairs

E4 (WE DO) $-5 + 8$

$8 - 5 = 3 \cdot + \text{has more: } 3$

E5 (WE DO) $6 + (-9)$

$9 - 6 = 3 \cdot - \text{has more: } -3$

12 $-7 + 10$

$10 - 7 = 3 \cdot + \text{has more: } 3$

13 $-9 + 4$

$9 - 4 = 5 \cdot - \text{has more: } -5$

14 $8 + (-12)$

$12 - 8 = 4 \cdot - \text{has more: } -4$

15 $-15 + 20$

$20 - 15 = 5 \cdot + \text{has more: } 5$

16 $25 + (-40)$

$40 - 25 = 15 \cdot - \text{has more: } -15$

17 $-36 + 50$

$50 - 36 = 14 \cdot + \text{has more: } 14$

Step 4: Three numbers

E6 (WE DO) $1 + (-4) + 5$
 $= -3 + 5 \cdot = 2$

18 $2 + (-5) + 4$
 $= -3 + 4 \cdot = 1$

19 $-3 + 6 + (-2)$
 $= 3 + (-2) \cdot = 1$

20 $12 + (-15) + (-6)$
 $= -3 + (-6) \cdot = -9$

21 $-20 + 8 + 15$
 $= -12 + 15 \cdot = 3$

Step 5: Adding in real life

E7 (WE DO) It is -5°C . The temperature rises 8°C . What is the temperature now?
 $-5 + 8 \cdot = 3^{\circ}\text{C}$

22 It is -4°C at dawn. By lunch it has warmed up 9°C . What is the temperature at lunch?
 $-4 + 9 \cdot = 5^{\circ}\text{C}$

23 A lift is at level -2 . It goes up 5 floors. Which level is it on now?
 $-2 + 5 \cdot = \text{level } 3$

24 Sam has \$15 in the bank. Sam spends \$40 using credit. What is the balance now?
 $15 + (-40) \cdot = -\$25$

25 A diver is at -18 m . She swims up 7 m, then down 4 m. Where is she now?
 $-18 + 7 + (-4) \cdot = -11 + (-4) \cdot = -15 \text{ m}$

Step 6: Is it right?

E8 (WE DO) Tom says $-6 + 2 = -8$. Is he right?
 $\text{start at } -6, \text{ move } 2 \text{ right} \cdot = -4 \cdot \text{No. The answer is } -4.$

26 Mia says $-8 + 5 = -13$. Is she right?
 $\text{start at } -8, \text{ move } 5 \text{ right} \cdot = -3 \cdot \text{No. The answer is } -3.$

27 Ben says $-7 + 3 + (-3) = -7$. Is he right?
 $3 + (-3) \text{ is a zero pair} \cdot -7 + 0 = -7 \cdot \text{Yes. Ben is right.}$

28 Ali says $-4 + \square = 2$ means $\square = -2$. Is he right?
from -4 to 2 is 6 right · $\square = 6$ · **No. $\square$ is 6, not -2 .**

29 Jo says adding always makes a number bigger. Is Jo right?
 $4 + (-6) = -2$ · -2 is smaller than 4 · No. It can get smaller.

1.05 Subtracting integers (booklet p. 11)

WE DO examples (teacher)

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|---|--|--|
| 1. Step 1, E1: $0 - 4$
$\rightarrow = -4$ | 9. Step 4, E1:
Minimum -9°C ,
maximum 9°C
$\rightarrow 9 - (-9) = 9 + 9 = 18^{\circ}\text{C}$ | 12. Step 6, E1: A submarine at -99 m rises 32 m. How deep is it now? $\rightarrow -99 + 32 = -67\text{ m}$ |
| 2. Step 1, E2: $11 - 12 \rightarrow -1$ | 10. Step 4, E2: From 11°C to $-9^{\circ}\text{C} \rightarrow -9 - 11 = -20^{\circ}\text{C}$, a fall of 20°C | 13. Step 6, E2: A bird is 87 m above sea level and a fish is 21 m below. How far apart are they? $\rightarrow 87 - (-21) = 87 + 21 = 108\text{ m}$ |
| 3. Step 1, E3: $0 - 15 \rightarrow -15$ | 11. Step 5, E1: $5 - (-11) + (-6) \rightarrow = 5 + 11 + (-6) = 16 + (-6) = 10$ | 14. Step 6, E3: What is 8 less than -12 ? $\rightarrow -12 - 8 = -20$ |
| 4. Step 2, E1: $-3 - 5 \rightarrow -8$ | | |
| 5. Step 2, E2: $-3 - 10 \rightarrow -13$ | | |
| 6. Step 2, E3: $-17 - 12 \rightarrow -29$ | | |
| 7. Step 3, E1: $-2 - (-6) \rightarrow = -2 + 6 = 4$ | | |
| 8. Step 3, E2: $-5 - (-7) \rightarrow = -5 + 7 = 2$ | | |

Questions

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|----------|-----------|--------------------------|
| 1. -7 | 10. -28 | 19. 10°C |
| 2. -5 | 11. -15 | 20. 13 |
| 3. -4 | 12. -19 | 21. -7 |
| 4. -6 | 13. -26 | 22. -14 |
| 5. -3 | 14. -36 | 23. 3 |
| 6. -4 | 15. -3 | 24. -4 |
| 7. -6 | 16. 10 | 25. -106 m |
| 8. -9 | 17. 17 | 26. 93 m |
| 9. -28 | 18. -4 | 27. -19 |

1.06 Multiplying integers (booklet p. 13)

WE DO examples (teacher)

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| 1. Step 1, E1: $2 \times (-3) \rightarrow (-3) + (-3) = -6$ | 10. Step 3, E3: $-10 \times (-6) \rightarrow 60$ | 15. Step 5, E2: The temperature drops 5°C each hour for 6 hours. What is the total change? $\rightarrow -5 \times 6 = -30^{\circ}\text{C}$ |
| 2. Step 1, E2: $5 \times (-7) \rightarrow -35$ | 11. Step 4, E1: $5 \times (-6) \times 2 \rightarrow = -30 \times 2 = -60$ | 16. Step 5, E3: A share price falls $\$5$ a week for 3 weeks. What is the total change? $\rightarrow -5 \times 3 = -\15 |
| 3. Step 1, E3: $3 \times (-5) \rightarrow -15$ | 12. Step 4, E2: $(-7) \times (-2) \times (-9) \rightarrow =$ | 17. Step 6, E1: Find the missing number: $-3 \times \square = -15 \rightarrow \square = -15 \div (-3) = 5$ |
| 4. Step 2, E1: $4 \times (-6) \rightarrow -$ | 13. Step 4, E3: $3 \times 9 \times 2 \rightarrow +$ | 18. Step 6, E2: Evaluate $(-7)^2$. $\rightarrow = -7 \times (-7) = 49$ |
| 5. Step 2, E2: $-8 \times 2 \rightarrow -$ | 14. Step 5, E1: A diver goes down 3 m every minute. How deep is she after 9 minutes? $\rightarrow -3 \times 9 = -27\text{ m}$ | |
| 6. Step 2, E3: $-6 \times (-8) \rightarrow 48$ | | |
| 7. Step 2, E4: $-4 \times 3 \rightarrow -12$ | | |
| 8. Step 3, E1: $2 \times (-9) = -18$, $1 \times (-9) = -9$, ... $\rightarrow 0 \times (-9) = 0$; $-1 \times (-9) = 9$; $-2 \times (-9) = 18$ | | |
| 9. Step 3, E2: $-8 \times (-7) \rightarrow 56$ | | |

19. Step 6, E3: Find two integers with a product of -35 and a sum of 2. $\rightarrow 7 \times (-5) = -35$; $7 + (-5) = 2$; 7 and -5

Questions

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|----------|--------------|---------------------------|
| 1. -12 | 15. -27 | 29. -40 |
| 2. -8 | 16. 60 | 30. 30 |
| 3. -4 | 17. -70 | 31. $+$ |
| 4. -24 | 18. 24 | 32. $+$ |
| 5. -8 | 19. 0, 5, 10 | 33. $+$ |
| 6. -32 | 20. 0, 7, 14 | 34. $-$ |
| 7. -30 | 21. 56 | 35. -18 m |
| 8. -54 | 22. 40 | 36. -10°C |
| 9. $-$ | 23. 88 | 37. $-\$30$ |
| 10. $+$ | 24. 132 | 38. -6 |
| 11. $+$ | 25. 72 | 39. 16 |
| 12. $-$ | 26. 42 | 40. 5 and -3 |
| 13. 14 | 27. 90 | 41. 3 |
| 14. 21 | 28. -72 | |

1.07 Dividing integers (booklet p. 15)

WE DO examples (teacher)

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| 1. Step 1, E1: $2 \times (-6) = -12 \rightarrow -12 \div 2 = -6$; $-12 \div (-6) = 2$ | 8. Step 3, E2: $14 \div \square = -2 \rightarrow \square = 14 \div (-2)$; $\square = -7$ | 12. Step 5, E2: The temperature fell 25°C over 5 hours. What was the change each hour? $\rightarrow -25 \div 5 = -5$; -5°C each hour |
| 2. Step 1, E2: $-60 \div 6 \rightarrow -10$ | 9. Step 4, E1: $-20 \div 5 \div (-2) \rightarrow = -4 \div (-2) = 2$ | 13. Step 5, E3: Find the average of these temperatures: -3°C , 1°C , -3°C , 1°C . $\rightarrow$ total: -4 ; $-4 \div 4 = -1$; average: -1°C |
| 3. Step 1, E3: $-32 \div 4 \rightarrow -8$ | 10. Step 4, E2: $\frac{35}{-5} \rightarrow = 35 \div (-5) = -7$ | 14. Step 6, E1: $-3 \times (-6) \div (-3) \rightarrow = 18 \div (-3) = -6$ |
| 4. Step 2, E1: $-10 \div (-5) \rightarrow 2$ | 11. Step 5, E1: A debt of $\$27$ is shared equally by 3 people. Write each share as an integer. $\rightarrow -27 \div 3 = -9$; each owes $\$9$: $-\$9$ | 15. Step 6, E2: $35 \div (-5) \times 3 \rightarrow = -7 \times 3 = -21$ |
| 5. Step 2, E2: $-14 \div (-7) \rightarrow 2$ | | |
| 6. Step 2, E3: $63 \div (-7) \rightarrow = -9$; check: $-9 \times (-7) = 63$ | | |
| 7. Step 3, E1: $\square \div (-6) = -3 \rightarrow \square = -3 \times (-6)$; $\square = 18$ | | |

Questions

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|--|-----------|--------------------------|
| 1. $-28 \div 4 = -7$; | 13. 5 | 29. -4 |
| $-28 \div (-7) = 4$ | 14. 3 | 30. -5 |
| 2. $-15 \div 3 = -5$; | 15. 2 | 31. 4 |
| $-15 \div (-5) = 3$ | 16. 9 | 32. -5 |
| 3. $-6 \div 3 = -2$; $-6 \div (-2) = 3$ | 17. -10 | 33. -8 |
| 4. $-14 \div 2 = -7$; | 18. 7 | 34. -6 |
| $-14 \div (-7) = 2$ | 19. 5 | 35. -7 |
| 5. -9 | 20. -2 | 36. $-\$30$ |
| 6. -4 | 21. 56 | 37. -6°C |
| 7. -3 | 22. -8 | 38. -12 |
| 8. -9 | 23. -35 | 39. -12 |
| 9. -2 | 24. 5 | 40. -8 |
| 10. -8 | 25. -15 | 41. 21 |
| 11. 8 | 26. -9 | 42. -12 |
| 12. -10 | 27. 49 | 43. -4 |
| | 28. 4 | |

1.08 Order of operations with integers

(booklet p. 17)

WE DO examples (teacher)

- Step 1, E1: $3 + (-9) \times (-9) \rightarrow \times$
- Step 1, E2: $-7 \times 8 - (-9) \rightarrow \times$
- Step 1, E3: $2 - (-6) \times (-4) \rightarrow = 2 - 24; = -22$
- Step 2, E1: $(-2 - (-4)) \times 2 \rightarrow = 2 \times 2; = 4$
- Step 3, E1: $-5 \times 3 + 16 \div (-4) \rightarrow = -15 + (-4); = -19$
- Step 4, E1: $15 \div 3 \times (-5) \rightarrow = 5 \times (-5); = -25$
- Step 4, E2: $-7 - 8 + 11 \rightarrow = -15 + 11; = -4$
- Step 5, E1: $-3 - (-4 + 8) \times 3 \rightarrow = -3 - 4 \times 3; = -3 - 12; = -15$
- Step 6, E1: Put brackets in to make it true: $-6 + 3 \times 5 = -15$
 $\rightarrow (-6 + 3) \times 5; = -3 \times 5; = -15$
- Step 6, E2: Mia says $-9 + 8 \times 5 = -5$. Is she right? Explain. $\rightarrow$
 $8 \times 5 = 40$ first; $-9 + 40 = 31$; No: the answer is 31

Questions

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|------------------|---------|-------------------------|
| 1. + in brackets | 13. 4 | 25. 20 |
| 2. + in brackets | 14. -12 | 26. -17 |
| 3. $\times$ | 15. 24 | 27. 6 |
| 4. $\div$ | 16. 30 | 28. -9 |
| 5. 22 | 17. -29 | 29. -20 |
| 6. -23 | 18. 15 | 30. -27 |
| 7. 8 | 19. -11 | 31. -4 |
| 8. 40 | 20. 6 | 32. -7 |
| 9. -14 | 21. -17 | 33. $(-7 + 9) \times 3$ |
| 10. -35 | 22. -10 | 34. No, it is 12 |
| 11. 12 | 23. 10 | 35. $(-3 + 7) \times 4$ |
| 12. 65 | 24. -5 | 36. No, it is 12 |

1.09 Integers on a calculator (booklet p. 19)

WE DO examples (teacher)

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| 1. Step 1, E1: $-83 + 12 \rightarrow (-) 83 + 12 =; = -71$ | 8. Step 4, E1: $(-34 - 6) \times (-4) \rightarrow$
bracket: -40 ;
$-40 \times (-4) = 160$ | 11. Step 6, E2: A submarine at -465 m rises 33 m a minute for 10 minutes. How deep is it now? $\rightarrow -465 + 33 \times 10; = -135$; it is at -135 m |
| 2. Step 1, E2: $-396 + (-140) \rightarrow -536$ | 9. Step 5, E1: $-33 \times 15 \rightarrow$ estimate:
$-30 \times 20 = -600$;
calculator:
-495 | 12. Step 6, E3: The lowest temperatures in Moscow for five days were -7°C , -20°C , -1°C , -9°C , -13°C . Find the average. $\rightarrow$
total: -50 ; $-50 \div 5 = -10$;
average: -10°C |
| 3. Step 1, E3: $-326 + 226 \rightarrow -100$ | 10. Step 6, E1: An account has \$200. A fee of \$63 is taken out every month for 8 months. What is the balance now?
$\rightarrow 200 - 63 \times 8$;
$= -304$; the balance is $-\$304$ | |
| 4. Step 2, E1: $-26 \times 14 \rightarrow -364$ | | |
| 5. Step 2, E2: $-533 \div (-41) \rightarrow 13$ | | |
| 6. Step 2, E3: $880 \div (-22) \rightarrow 880 \div ((-) 22) =; = -40$ | | |
| 7. Step 3, E1: $-34 \times 29 \rightarrow$ sign: $-$ (different signs); $= -986$ | | |

Questions

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| 1. -152 | 2. -78 | 3. -100 |
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| 4. -108 | 16. 15 | 28. -2 751 |
| 5. -349 | 17. 540 | 29. 322 |
| 6. 82 | 18. 640 | 30. -1 155 |
| 7. -499 | 19. -28 | 31. -1 562 |
| 8. -455 | 20. -2 494 | 32. ≈ -400 ; -494 |
| 9. -57 | 21. 1 539 | 33. $\approx -1\ 200$; $-1\ 184$ |
| 10. -413 | 22. 1 406 | 34. ≈ -900 ; -891 |
| 11. -1 476 | 23. -1 020 | 35. $\approx -1\ 200$; $-1\ 037$ |
| 12. -5 | 24. 945 | 36. ≈ -900 ; $-1\ 054$ |
| 13. -1 364 | 25. -1 794 | 37. $\approx -3\ 200$; $-2\ 905$ |
| 14. -45 | 26. 1 962 | 38. $-\$217$ |
| 15. -1 160 | 27. -187 | 39. -223 m |

1.10 Integer problems (booklet p. 21)

WE DO examples (teacher)

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|---|---|--|
| 1. Step 1, E1: -11°C , rises $16^\circ\text{C} \rightarrow 5^\circ\text{C}$ | 7. Step 3, E2: How many floors does a lift travel from level -4 to level 4 ? $\rightarrow 4 - (-4); = 4 + 4; = 8$ floors | 11. Step 5, E3: A team scores $+2$ for a win and -1 for a loss. It wins 3 games and loses 6. What is its score? $\rightarrow 3 \times 2 = 6$; $6 \times (-1) = -6$; $6 + (-6) = 0$ |
| 2. Step 1, E2: -4°C , falls $12^\circ\text{C} \rightarrow -16^\circ\text{C}$ | 8. Step 4, E1: Balance after transaction 1 $\rightarrow 60 - 85; = -\25 | 12. Step 6, E1: Kai gets 2 correct and 5 wrong. What is Kai's score? $\rightarrow 2 \times 3 = 6$; $5 \times (-1) = -5$; $6 + (-5) = 1$ |
| 3. Step 1, E3: from -6°C to $9^\circ\text{C} \rightarrow 15^\circ\text{C}$ | 9. Step 5, E1: A diver is at -70 m. She rises 3 m a minute for 2 minutes. Where is she now? $\rightarrow 3 \times 2 = 6$; $-70 + 6; = -64$ m | 13. Step 6, E2: Lea answers all 10 questions and scores 2. How many did she get correct? $\rightarrow$ try 3: $3 \times 3 = 9$; 7 wrong: $9 - 7 = 2$; 3 correct |
| 4. Step 1, E4: from -5°C to $11^\circ\text{C} \rightarrow 16^\circ\text{C}$ | 10. Step 5, E2: It is -6°C at 6 a.m. It warms 5°C an hour for 5 hours. What is the temperature now? $\rightarrow 5 \times 5 = 25$; $-6 + 25; = 19^\circ\text{C}$ | |
| 5. Step 2, E1: the cliff top and the diver $\rightarrow 40 - (-15); = 55$ m | | |
| 6. Step 3, E1: A lift is at level 3. It goes down 4 floors. Which level is it on now? $\rightarrow 3 - 4; =$ level -1 | | |

Questions

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|------------------------|----------------|------------------------|
| 1. -15°C | 11. 60 m | 21. $-\$35$ |
| 2. -5°C | 12. 40 m | 22. $\$18$ |
| 3. -6°C | 13. 25 m | 23. $-\$57$ |
| 4. 12°C | 14. 35 m | 24. -28 m |
| 5. -6°C | 15. 80 m | 25. 15°C |
| 6. -6°C | 16. level -4 | 26. 3 |
| 7. 9°C | 17. 11 floors | 27. 10 correct |
| 8. 3°C | 18. level 0 | 28. 11 |
| 9. 17°C | 19. 4 floors | |
| 10. 21°C | 20. $-\$272$ | |